

Harsh Karia

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Education

University of California, Davis, Bachelors of Science in Computer Science Sep 2022 – Dec 2025

- **Relevant Coursework:** Machine Learning, Deep Learning, Computer Networks, Data Structures and Algorithms, Operating Systems, UNIX Development, Statistics, Probability, LLMs, Cybersecurity, Cryptography, AI

Skills

Technical: Python, C++ , Rust, Go, React, JavaScript, TypeScript, Machine Learning, Deep Learning, Backend Development, GCP, Linux, Operating Systems, Infrastructure, Docker, LLMs, PyTorch, TensorFlow , Pandas, NumPy

Experience

Ethereum Foundation Jun 2025 – Sep 2025

AI and Security Engineering Intern

- Currently focused on **open-source** code **security** and blockchain **infrastructure**, automating code reviews of over **1 million lines of code** and 200+ engineers using **C++ , Python, Go, Rust, and GitHub Actions**
- **Scoped and led** custom **abstract syntax trees + RAG** system project to represent syntactic structure of code and using **machine learning** for fuzzy matching Consensus layer specs to **identify vulnerabilities 80% faster**

University of California, Davis Research Jan 2025 – Dec 2025

Machine Learning Research Intern

- **Researching LLM memory** behavior and data persistence under guidance from Prof. Zubair Shafiq, with a focus on **developing first-of-its-kind benchmarks** for long-term memory evaluation
- Observing **memory of LLMs and MCP** to identify lapses in data deletion, personalization, **vulnerabilities**, and commercial memory controls in **C++ and Python** from dataset of **200,000 conversations**

Corgi AI (YC Startup) Sep 2024 – Oct 2024

Software Engineering Intern

- Introduced automations for **financial data parsing** and email marketing using OpenAI's **LLM** models and **built backend APIs** for **financial data modeling** with **Python and React**, reducing employee requirements by **100x**
- Developed insurance policy pricing by building in-house **risk management algorithms** and **created databases** for property managers using **PostgreSQL and Python** for **FastAPI backend**

American Wild Horse Campaign June 2023 – Sept 2023

Software Engineering Intern

- Spearheaded end-to-end development of a **secure full-stack mobile application** using **React** and **JavaScript**, facilitating **data collection** and analysis with **Pandas and NumPy** for **10,000+** horse riders
- Designed and deployed frontend user interfaces using **Android UI** and integrated **object detection models** with **95% accuracy** to automate wild horse identification in images using Azure Cloud and **TypeScript backend**

Projects

AVIA (Stanford University Hackathon Winner) [GitHub](#)

- Implemented **full-stack** AI command center that consolidates **real-time biometric data** for **11 vital metrics** using Terra API and PPG recordings, **predicting blackouts** and pilot failures with SVMs and neural nets
- Created live strategy optimizations using **Python backend** with **React, JavaScript, and Tailwind CSS** frontend

CourtCheck [GitHub](#)

- Engineered **machine learning based tennis analysis** tool using **Python, PyTorch, OpenCV, Pandas**, and **Detectron2** to achieve **95% accuracy** in court and ball detection, **reducing human error** in line calls by **90%**
- Utilized **NVIDIA T4 GPUs** to train Detectron2 ML models with **ResNet-50 backbone** and **Python backend**, processing **10,000 images** from tennis video film for keypoint detection with **TypeScript dashboard** analytics

Volare [GitHub](#)

- Built **backend pipeline** for realtime interview prep tool **dockerized Python, Next.js/TypeScript frontend**, and **Supabase**, **deployed to cloud**, reducing end-to-end time by **40%** and increasing upload **success to 99%**
- Integrated **Hume AI, Whisper, Google Gemini**, and **Groq AI models** to analyze **48 user emotions** and 36 facial expressions, orchestrating **LLM prompts** and classifiers to generate real-time interview feedback

MMA Pose Estimation [GitHub](#)

- Developed an MMA pose estimation and analysis tool and collected, annotated, and trained on **custom dataset** using **14 keypoint skeleton** mapping structure with **CNNs**
- streamlined user experience with **real-time angle calculations**, enabling **automated feedback** on key movements and injury-prone postures with **TypeScript**